

## BEAVER ISLAND, MI, USA

WMO: 722186

Lat: 45.689N

Lon: 85.566W

Elev: 203

StdP: 98.91

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 04892

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -18.4      | -15.7 | -22.4                         | 0.5 | -17.5 | -19.7 | 0.7 | -14.8 | 11.4                  | -3.0 | 10.6 | -3.4 | 2.0                   | 40   | 0.582 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.3                   | 27.8            | 20.4 | 26.5 | 19.6 | 25.2 | 18.9 | 21.9                | 25.9 | 21.1 | 24.6 | 20.3 | 23.7 | 4.2                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.7                            | 15.7 | 24.1 | 19.9 | 15.0 | 23.3 | 19.0 | 14.1 | 22.4 | 65.0          | 26.0 | 62.0 | 24.7 | 59.2 | 23.7 | 24.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 10.0          | 8.7  | 7.8 | -23.0                      | 30.8 | 4.6                | 1.1 | -26.3                                              | 31.6 | -29.0      | 32.2 | -31.6      | 32.8 | -35.0      | 33.6 |    |    |
| (5)               |      |     | -23.3                      | 23.3 | 4.5                | 0.8 | -26.5                                              | 23.9 | -29.2      | 24.3 | -31.7      | 24.8 | -34.9      | 25.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 7.0    | -5.3  | -5.9  | -1.5  | 4.5   | 10.8  | 16.0  | 19.1  | 19.0  | 15.6  | 9.3   | 3.3   | -1.9  |
|                                                                                  | DBStd       | 9.91   | 5.42  | 5.40  | 5.71  | 4.25  | 4.34  | 3.53  | 3.09  | 2.97  | 3.91  | 4.42  | 4.23  | 4.44  |
|                                                                                  | HDD10.0     | 2137   | 474   | 444   | 359   | 171   | 41    | 2     | 0     | 0     | 3     | 69    | 204   | 370   |
|                                                                                  | HDD18.3     | 4280   | 732   | 678   | 616   | 414   | 236   | 88    | 29    | 27    | 100   | 283   | 450   | 628   |
|                                                                                  | CDD10.0     | 1037   | 0     | 0     | 2     | 7     | 67    | 181   | 282   | 279   | 170   | 46    | 3     | 0     |
|                                                                                  | CDD18.3     | 139    | 0     | 0     | 0     | 0     | 3     | 17    | 52    | 48    | 17    | 2     | 0     | 0     |
|                                                                                  | CDH23.3     | 817    | 0     | 0     | 1     | 0     | 41    | 118   | 335   | 244   | 75    | 4     | 0     | 0     |
|                                                                                  | CDH26.7     | 107    | 0     | 0     | 0     | 0     | 4     | 18    | 54    | 23    | 7     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 3.5    | 4.0   | 3.8   | 3.5   | 3.7   | 3.3   | 2.9   | 2.9   | 2.9   | 3.1   | 3.8   | 4.1   | 4.3   |
| Precipitation                                                                    | PrecAvg     | 843    | 68    | 42    | 52    | 68    | 76    | 69    | 71    | 71    | 100   | 93    | 71    | 71    |
|                                                                                  | PrecMax     | 995    | 135   | 72    | 121   | 158   | 133   | 168   | 137   | 138   | 218   | 170   | 135   | 160   |
|                                                                                  | PrecMin     | 707    | 14    | 8     | 7     | 20    | 29    | 24    | 27    | 16    | 25    | 37    | 24    | 10    |
|                                                                                  | PrecStd     | 81     | 34    | 20    | 35    | 37    | 28    | 38    | 30    | 31    | 45    | 41    | 32    | 38    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 7.4   | 7.5   | 19.2  | 20.9  | 27.3  | 28.9  | 30.3  | 28.9  | 27.6  | 23.0  | 17.2  | 11.3  |
|                                                                                  |             | MCWB   | 5.6   | 5.6   | 14.0  | 12.4  | 18.6  | 20.6  | 20.9  | 21.4  | 20.5  | 17.8  | 12.6  | 10.0  |
|                                                                                  | 2%          | DB     | 4.9   | 4.1   | 12.2  | 17.1  | 23.9  | 26.3  | 27.9  | 27.2  | 25.5  | 21.0  | 13.6  | 7.0   |
|                                                                                  |             | MCWB   | 3.5   | 2.6   | 8.3   | 10.2  | 16.6  | 19.0  | 20.5  | 20.6  | 19.5  | 17.0  | 10.9  | 5.6   |
|                                                                                  | 5%          | DB     | 2.9   | 2.4   | 8.0   | 14.1  | 21.3  | 24.5  | 26.6  | 26.0  | 23.8  | 18.5  | 11.3  | 5.4   |
|                                                                                  |             | MCWB   | 1.8   | 1.2   | 5.2   | 8.4   | 14.7  | 17.7  | 19.6  | 19.6  | 18.9  | 15.1  | 8.9   | 4.0   |
|                                                                                  | 10%         | DB     | 1.9   | 1.2   | 5.9   | 12.1  | 18.8  | 22.7  | 25.2  | 24.8  | 22.4  | 16.4  | 9.9   | 3.9   |
|                                                                                  |             | MCWB   | 1.1   | 0.1   | 3.0   | 7.0   | 13.0  | 16.6  | 19.0  | 19.0  | 18.1  | 13.4  | 7.6   | 2.6   |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 6.4   | 6.0   | 14.7  | 13.9  | 19.7  | 22.2  | 23.1  | 22.9  | 21.9  | 19.9  | 13.3  | 10.1  |
|                                                                                  |             | MCDB   | 7.3   | 7.4   | 19.0  | 18.6  | 25.1  | 27.1  | 27.7  | 26.7  | 25.6  | 22.7  | 16.1  | 11.0  |
|                                                                                  | 2%          | WB     | 3.6   | 3.1   | 8.7   | 11.1  | 17.9  | 20.3  | 21.8  | 21.9  | 20.9  | 17.6  | 11.1  | 5.8   |
|                                                                                  |             | MCDB   | 4.8   | 4.4   | 11.1  | 15.7  | 22.4  | 24.6  | 26.0  | 25.7  | 23.6  | 19.6  | 13.3  | 6.7   |
|                                                                                  | 5%          | WB     | 2.1   | 1.4   | 5.2   | 9.1   | 15.8  | 19.0  | 20.8  | 21.1  | 19.9  | 15.4  | 9.2   | 4.2   |
|                                                                                  |             | MCDB   | 3.0   | 2.2   | 8.1   | 13.4  | 20.1  | 23.1  | 24.6  | 24.1  | 22.7  | 18.2  | 11.1  | 5.4   |
|                                                                                  | 10%         | WB     | 1.0   | 0.2   | 3.2   | 7.4   | 13.9  | 17.7  | 20.1  | 20.2  | 18.8  | 13.7  | 7.8   | 2.8   |
|                                                                                  |             | MCDB   | 1.7   | 0.9   | 5.4   | 11.1  | 17.6  | 21.5  | 23.8  | 23.3  | 21.7  | 16.4  | 9.3   | 3.9   |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.6   | 6.6   | 8.0   | 9.0   | 11.0  | 10.7  | 10.3  | 9.6   | 9.5   | 7.3   | 5.8   | 4.9   |
|                                                                                  |             | MCDBR  | 5.8   | 6.7   | 10.5  | 13.5  | 14.3  | 12.5  | 11.6  | 10.4  | 10.0  | 9.6   | 8.4   | 6.4   |
|                                                                                  |             | MCWBR  | 5.5   | 5.9   | 7.4   | 8.1   | 7.9   | 6.6   | 5.8   | 5.5   | 5.3   | 6.4   | 6.5   | 6.1   |
|                                                                                  | 5% WB       | MCDBR  | 5.5   | 6.5   | 10.4  | 12.2  | 12.7  | 10.7  | 9.6   | 8.6   | 8.6   | 8.8   | 8.0   | 6.5   |
|                                                                                  |             | MCWBR  | 5.3   | 6.0   | 7.6   | 8.3   | 8.0   | 6.5   | 5.5   | 5.1   | 5.8   | 6.6   | 6.6   | 6.2   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.285  | 0.290 | 0.315 | 0.352 | 0.381 | 0.392 | 0.389 | 0.383 | 0.368 | 0.339 | 0.305 | 0.280 |       |
|                                                                                  | taud        | 2.458  | 2.404 | 2.399 | 2.360 | 2.361 | 2.369 | 2.369 | 2.405 | 2.419 | 2.444 | 2.495 | 2.525 | 2.511 |
|                                                                                  | Ebn at Noon | 831    | 896   | 918   | 906   | 887   | 875   | 874   | 866   | 849   | 828   | 802   | 803   |       |
|                                                                                  | Edh at Noon | 68     | 87    | 101   | 115   | 119   | 119   | 113   | 108   | 97    | 80    | 64    | 59    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.15   | 2.08  | 3.49  | 4.60  | 5.65  | 6.12  | 6.16  | 5.23  | 3.97  | 2.31  | 1.23  | 0.85  |       |
|                                                                                  | RadStd      | 0.13   | 0.21  | 0.36  | 0.48  | 0.47  | 0.44  | 0.25  | 0.33  | 0.32  | 0.27  | 0.17  | 0.08  |       |

## Historical Trends

|                              |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                          | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| (46) Station Only            | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     |         |         |
| (47) Regional (43 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     |         |         |

Nomenclature: See separate page